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1 Chapter 01 — RL Basics: Implementation Report

Environment: April 2026

Algorithms: DQN (CartPole-v1), PPO (LunarLander-v3)

Targets: DQN ≥ 475 , PPO ≥ 200 (avg over 100 episodes)

Status: Both targets achieved ✓

1.1 Table of Contents

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1.2 Overview

Chapter 01 implements two foundational RL algorithms from scratch in PyTorch, with extensive inline comments explaining the *why* behind each design choice. Both algorithms are compared

against Stable-Baselines3 (SB3): PPO with our hyperparameters, DQN with the RL Baselines3 Zoo’s tuned CartPole settings (see Comparison).

Source layout:

```
implementation/step01/
├─ dqn/
│   ├── replay_buffer.py    # Circular buffer, random batch sampling
│   ├── q_network.py        # MLP Q-function
│   ├── agent.py            # DQNAgent: select/store/train/sync/save
│   └─ train.py             # Training loop, best-model checkpointing, early stop
├─ ppo/
│   ├── networks.py         # PolicyNetwork (actor), ValueNetwork (critic)
│   ├── gae.py              # Generalized Advantage Estimation (reverse sweep)
│   ├── agent.py            # PPOAgent: rollout → GAE → clipped SGD
│   └─ train.py             # Training loop, rollout collection, logging
├─ compare_sb3.py           # Comparison script - reads TB logs, trains SB3
├─ config.py                # DQN_CONFIG, PPO_CONFIG
└─ utils/
    └─ logger.py            # TensorBoard SummaryWriter wrapper
```

1.3 DQN — Deep Q-Network

1.3.1 Architecture

Environment: CartPole-v1 — 4-dimensional continuous observation, 2 discrete actions, maximum 500 steps per episode.

Q-network: Two hidden layers, ReLU activations, no output activation (Q-values can be any real number). Size [obs_dim=4 → 128 → 128 → 2].

Replay buffer: Pre-allocated numpy arrays with a circular write cursor. Stores (state, action, reward, next_state, done) tuples. Random mini-batch sampling breaks temporal correlation between consecutive transitions.

Target network: A frozen copy of the Q-network, synced every `target_update_freq` episodes. Prevents the unstable feedback loop where the TD target moves at the same rate as the predictions being trained.

1.3.2 Key Design Decisions

Decision	Choice	Rationale
Exploration schedule	Episode-based ϵ -decay ($\epsilon \times 0.995$ per episode)	More predictable than step-based when episode length varies
Minimum epsilon	$\epsilon_{\min} = 0.001$	Avoids random action deaths after convergence; lower than SB3's default 0.05
Target sync	Every 5 episodes	Balances stability vs. speed; too infrequent (10) caused slower plateaus
Network size	[128, 128]	[64, 64] plateaued around 300 avg reward; larger network learned faster
Buffer size	50,000	Smaller (10K) caused over-fitting to recent experience

1.3.3 Hyperparameter Iterations

The final configuration was reached after 3 tuning rounds:

Run 1 — Baseline (500 episodes)

- Config: `buffer=10K, hidden=[64,64], episodes=500, target_freq=10,  $\epsilon_{\min}=0.01$`
- Result: Final avg 225, never crossed 475 target
- Problem: Small buffer and network; infrequent target updates

Run 2 — First tuning (1000 episodes)

- Config: `buffer=50K, hidden=[128,128], episodes=1000, target_freq=5,  $\epsilon_{\min}=0.01$`
- Result: Peak avg **479.1** at episode 810, but degraded to 302.6 by end
- Problem: $\epsilon_{\min}=0.01$ means 1% random actions permanently — enough to kill long episodes
- Fix needed: Lower $\epsilon_{\min}$ + add best-model checkpointing

Run 3 — Final (1500 episodes, early stopping added)

- Config: same + $\epsilon_{\min}=0.001$, `episodes=1500`
- Added best-model saving and early stop (saves when 475 target is first hit, stops training)
- Result: **Solved at episode 1011**, `Avg(100) = 477.5 ✓`

1.3.4 Final Results

Metric	Value
Solved at episode	1011
Final avg reward (100 eps)	477.5
Target	475.0
Total episodes trained	1011 (early stop)
Best model	<code>models/dqn_cartpole_best.pt</code>

1.4 PPO — Proximal Policy Optimization

1.4.1 Architecture

Environment: LunarLander-v3 — 8-dimensional continuous observation, 4 discrete actions. A successful landing scores +200; crash is −100; fuel usage is penalised.

Actor-Critic (separate networks): - **PolicyNetwork:** [obs=8 → 128 → 128 → 4] with logits → Categorical distribution - **ValueNetwork:** [obs=8 → 128 → 128 → 1] with unbounded scalar output

Separate networks avoid gradient interference between the policy and value objectives. A single Adam optimizer trains both via a combined loss.

GAE (Generalized Advantage Estimation, $\lambda=0.95$): Computed by a reverse sweep over the rollout buffer. The (1 - done) mask zeroes out the bootstrap across episode boundaries, correctly handling rollouts that span multiple episodes.

Clipped surrogate objective:

$$L^{CLIP} = \mathbb{E} \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where $r_t(\theta) = \pi_\theta(a|s)/\pi_{old}(a|s)$ and $\epsilon = 0.2$.

1.4.2 Key Design Decisions

Decision	Choice	Rationale
Advantage normalisation	Zero mean, unit std per rollout	Stabilises gradient magnitude across diverse reward scales
Entropy bonus	<code>coef = 0.01</code>	Prevents premature policy collapse; encourages exploration

Decision	Choice	Rationale
Gradient clipping	<code>max_norm = 0.5</code>	Avoids exploding gradients during early training
Rollout length	<code>n_steps = 2048</code>	Long enough for multi-step credit assignment across landings
Mini-batch SGD	10 epochs, batch=64	Standard PPO; re-using each rollout 10× before discarding

1.4.3 Hyperparameter Iterations

Run 1 — Baseline (300K steps, hidden=[64,64])

- Result: Peak avg **179.9**, never crossing 200 target
- Problem: Small network underfitted LunarLander’s 8-dim observation
- Also: 300K steps just barely too short for full convergence

Run 2 — Final (500K steps, hidden=[128,128])

- Network size doubled; training budget increased by 67%
- Result: **Solved at 264K steps** (early stop triggered), Avg(100) = **202.2** ✓
- Notably: solved 236K steps *before* the training budget ran out

1.4.4 Final Results

Metric	Value
Solved at step	264,192
Final avg reward (100 eps)	202.2
Target	200.0
Total steps trained	264,192 (early stop)
Best model	<code>models/ppo_lunarlander_best.pt</code>

1.5 Comparison with SB3 Baselines

SB3 PPO was trained with our core hyperparameters (learning rate, γ , network size, clip range, etc.) and the same 500K-step budget. SB3 DQN was trained with the RL Baselines3 Zoo’s tuned CartPole-v1 hyperparameters for 100K steps (run of 6 April 2026, seed 42). Source: `implementation/step01/compare_sb3.py`; the SB3 episode rewards are in `sb3_results_cache.json`.

1.5.1 4.1 DQN on CartPole-v1

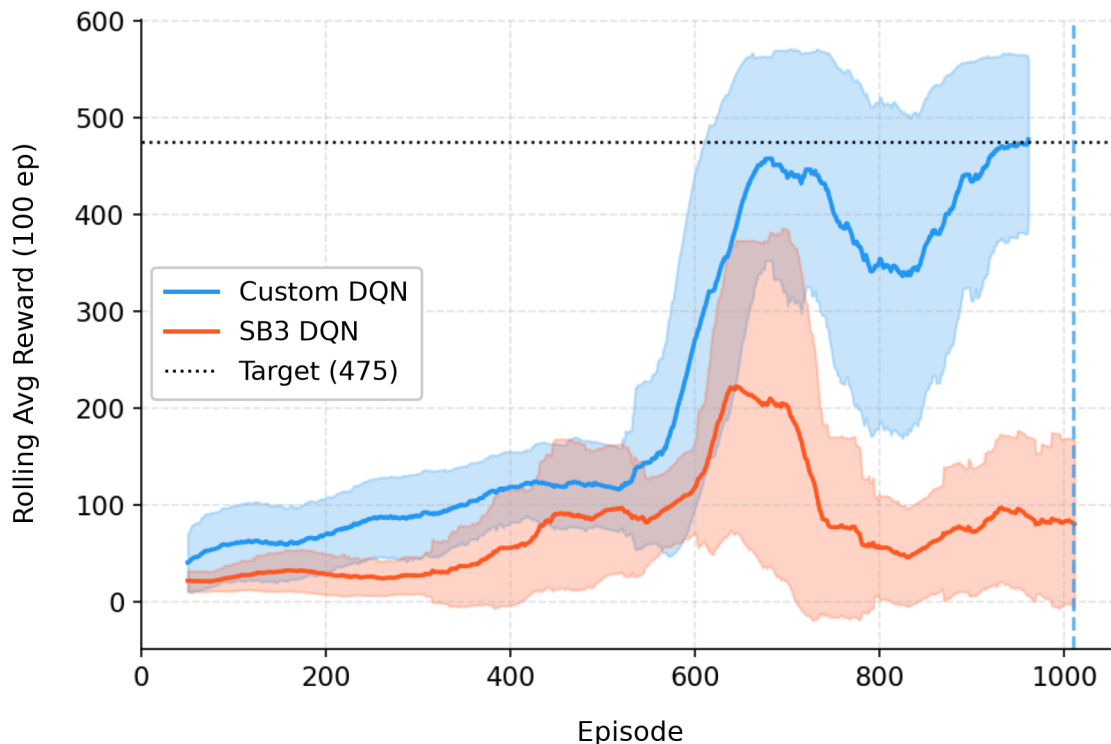


Figure 1: DQN on CartPole-v1: rolling 100-episode average of the reward, our implementation against SB3 (first 1,061 of SB3’s 1,255 episodes).

Custom DQN solves CartPole by episode ~ 1011 and hits the 475 target (avg over 100 eps). **SB3 DQN** (RL Zoo settings: learning rate 0.0023, [256, 256], target sync every 10 steps, `train_freq`=256 with 128 gradient steps, $\epsilon \rightarrow 0.04$ over the first 16K steps) ran 100K steps (1,255 episodes). Its best rolling-100 average is **222.4**, at episode 695 (≈ 51 K steps); it ends at 134.3. The figure shows its first 1,061 episodes.

Superseded run. An earlier SB3 DQN run (3 April 2026: 750K steps, 17,176 episodes, our core hyperparameters with `exploration_fraction`=0.36 and `target_update_interval`=1000) peaked at 293.9. The run above replaced it on 6 April; its results remain in the git history of `sb3_results_cache.json` (commit 4a173d2).

Why the gap? Not established. Because SB3 ran with tuned rather than matched settings, the comparison is not like-for-like. Two differences are untested:

1. **Exploration during the plotted episodes:** the rewards are those of training episodes, in which SB3 still acts at random 4% of the time (final $\epsilon = 0.04$), while our ϵ ($\times 0.995$ per episode) was about 0.006 when the task was solved.

2. **Updates per step:** SB3 makes 128 gradient steps every 256 environment steps, half as many updates per step as our agent, which updates on every step.

Each implementation was run once (ours without a fixed seed), and the Zoo trains these settings for 50K steps, so SB3’s shortfall at 100K needs a second seed before it is read as a property of SB3.

1.5.2 4.2 PPO on LunarLander-v3

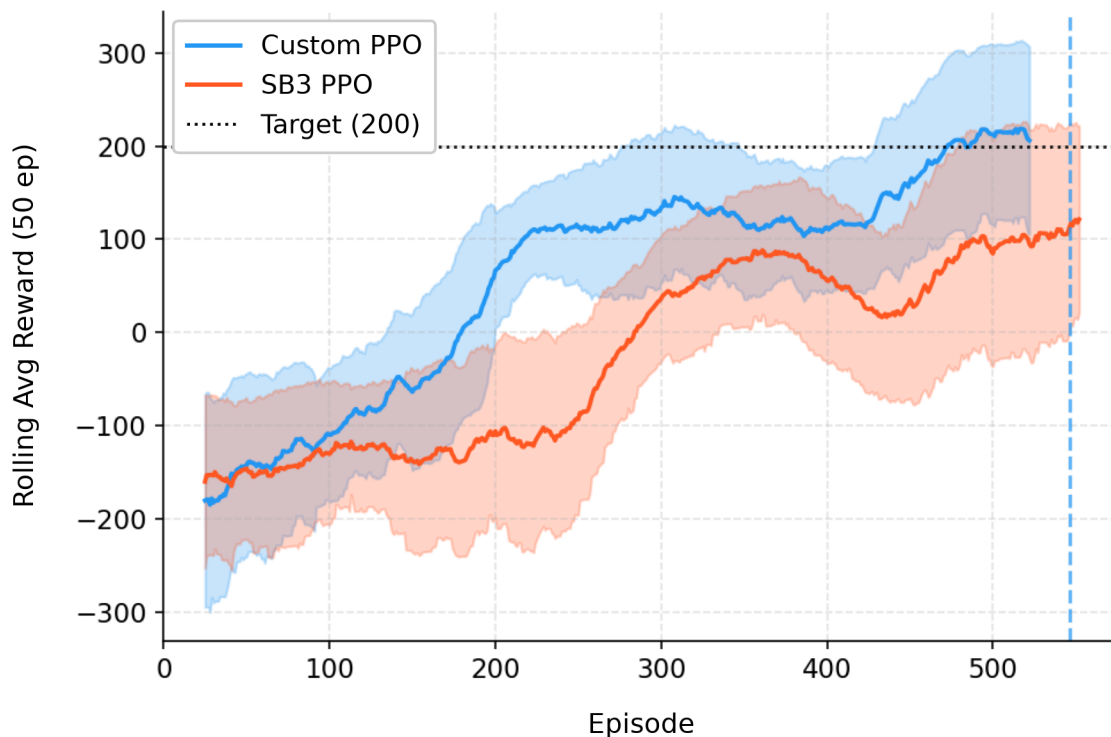


Figure 2: PPO on LunarLander-v3: rolling 50-episode average of the reward, our implementation (final run) against SB3 (first 577 of its 1,224 episodes).

Custom PPO crosses the 200 target by episode ~543 (264K steps). **SB3 PPO** with the same step budget (500K steps) peaks at a best rolling-100 of **131.2** and is still climbing at the end of training. (The figure’s rolling window is 50 episodes; the numbers in the text use 100.)

The gap is narrower here than in DQN. Both use identical PPO hyperparameters. The remaining difference may come from: - SB3’s orthogonal weight initialisation; - SB3 normalising advantages per mini-batch rather than per rollout; - chance: each implementation was run once.

Given more training budget (e.g. 1–2M steps), SB3 PPO would likely converge to target. In this single run, our custom PPO used its samples more efficiently than SB3 on this task.

1.5.3 4.3 Peak Performance Summary

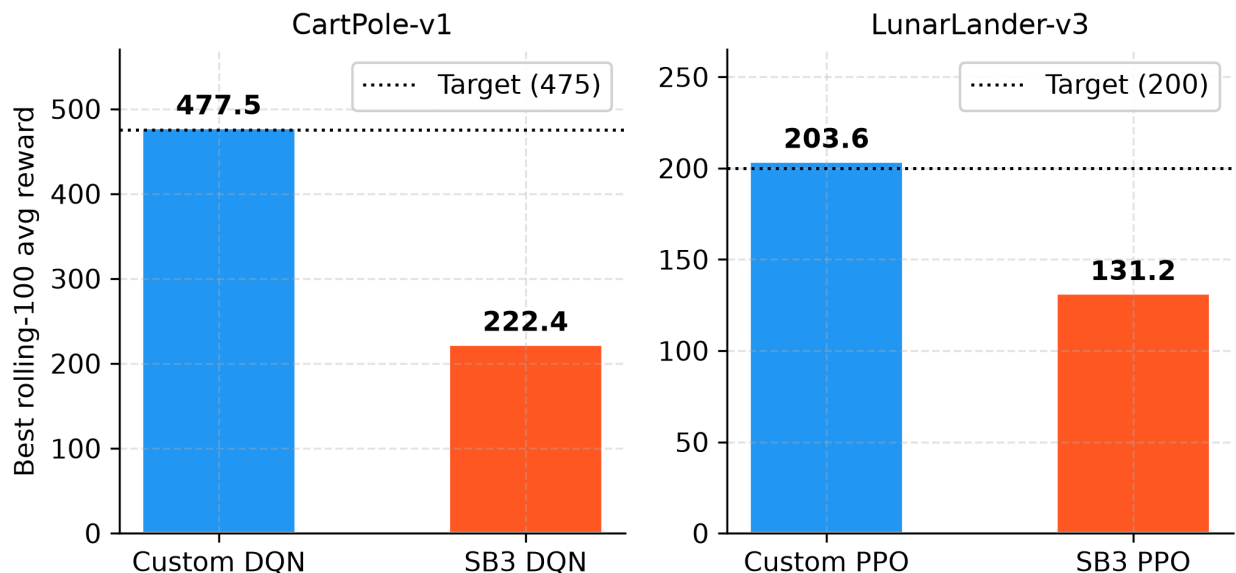


Figure 3: Best rolling 100-episode average: our implementations against SB3.

Algorithm	Environment	Our Best Avg (rolling 100)	SB3 Best Avg (rolling 100)	Target
DQN	CartPole-v1	477.5	222.4	475
PPO	LunarLander-v3	203.6	131.2	200

Methodology note: “Best rolling-100 avg” is the maximum of the 100-episode moving average over the entire training curve. This metric is fair regardless of early stopping: it measures peak capability, not where training happened to end. Step budgets: PPO 500K for both; SB3 DQN 100K (our DQN stopped at episode 1011). Our values come from the final runs’ logs as read on 3 April 2026 (the logs themselves are not kept); SB3’s are computed from `sb3_results_cache.json`.

1.6 Key Learnings

1.6.1 5.1 DQN

1. **Best-model checkpointing is essential:** DQN is prone to catastrophic forgetting once epsilon hits its minimum. Saving the best model prevents losing a good solution.
2. **$\epsilon_{\min}$ matters more than expected:** Dropping from 0.01 to 0.001 was the difference between a degrading policy and a stable one.

3. **Target network sync frequency:** Every 5 episodes was consistently better than 10. Too infrequent $\rightarrow$ slow learning; too frequent $\rightarrow$ instability (equivalent to no target network).
4. **Episode-based vs step-based exploration:** For environments with growing episode length (like CartPole), episode-based decay is more natural — the agent sees more of the environment’s dynamics before stopping exploration.

1.6.2 5.2 PPO

1. **Network size bottleneck:** [64, 64] bottlenecked learning on LunarLander’s 8-dim space. Doubling to [128, 128] was the decisive change.
2. **Advantage normalisation was kept throughout:** without it, gradient magnitudes would scale with absolute rewards, which in LunarLander range from -500 to $+300$. No run without it was recorded, so its effect here is expected rather than measured.
3. **GAE $\lambda = 0.95$ is robust:** The standard value worked well without tuning.
4. **Entropy bonus prevents stagnation:** Without it, the policy converged to “hover in place” (a locally safe but suboptimal strategy) early in LunarLander training.

1.6.3 5.3 Implementation vs SB3

- A clean from-scratch implementation is 300–500 lines vs SB3’s $\sim 10\text{K}$ lines. The tradeoff: fewer features but complete transparency and debuggability.
- The SB3 DQN baseline used the RL Zoo’s tuned CartPole settings and, in a single run at 100K steps, peaked at 222.4; this needs a second seed before it is read as a property of SB3. Our task-specific implementations reached both targets.
- For research purposes (thesis work on opponent exploitation, Chapters 7–8), custom implementations allow surgical modifications (e.g. injecting belief-state observations) that would require deep SB3 subclassing.

1.7 Appendix — Reproduction

From repo root, with .venv activated:

Train DQN (early stops ~episode 1000):

python implementation/step01/dqn/train.py

Train PPO (early stops ~step 264K):

python implementation/step01/ppo/train.py

Regenerate comparison figures (caches SB3 results after first run):

```
python implementation/step01/compare_sb3.py
```

```
# View training curves in TensorBoard:
```

```
tensorboard --logdir implementation/step01/logs/
```

The TensorBoard logs of the custom runs are not kept in the repository, so `compare_sb3.py` cannot redraw the learning curves until the runs are repeated. The figures here are the saved renders (DQN: 6 April 2026; PPO: 3 April 2026); `deliverables/reports/step01/summary/make_comparison_panels.py` crops their rolling-average panels.

Generated figures are in `deliverables/reports/step01/figures/`.